

The diminished effects of gravity in space (ie, microgravity) can alter the function and regulation of body systems, including those involved in movement and fluid distribution. In addition to the reduced loads placed on muscle and bone, some of the fluid volume that normally collects in the lower extremities is shifted toward the head.

One problem encountered by astronauts as a result of their exposure to microgravity is orthostatic hypotension, a reduced ability to maintain blood pressure when assuming an upright posture. When orthostatic hypotension manifests, it typically occurs during descent back to Earth and shortly thereafter.

Opportunities for experimentation in space are limited and costly. Therefore, experimental scenarios intended to model one or more aspects of spaceflight have been developed to allow research relevant to astronaut performance and health to be conducted on Earth. One such model is bed rest, which induces a fluid shift toward the head and alters muscle activity levels.

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If nonexercise contractile activity in postural muscles is decreased during spaceflight, but Ca^{2+} exchange with the extracellular fluid is unaffected, which table shows the expected changes in myosin ATPase and sarcoplasmic reticulum Ca^{2+} -ATPase (SERCA) activities in those muscles? (Note: L = lower rate of ATP hydrolysis, H = higher rate of ATP hydrolysis.)

☐ A.

Condition	ATPase	Rate of ATP hydrolysis
On earth	Myosin ATPase	L
	Sarcoplasmic reticulum Ca^{2+} -ATPase	L
Spaceflight	Myosin ATPase	H
	Sarcoplasmic reticulum Ca^{2+} -ATPase	H

[3%]

☐ B.

Condition	ATPase	Rate of ATP hydrolysis
On earth	Myosin ATPase	H
	Sarcoplasmic reticulum Ca^{2+} -ATPase	L
Spaceflight	Myosin ATPase	L
	Sarcoplasmic reticulum Ca^{2+} -ATPase	L

[10%]

☐ C.

Condition	ATPase	Rate of ATP hydrolysis
On earth	Myosin ATPase	H
	Sarcoplasmic reticulum Ca^{2+} -ATPase	H
Spaceflight	Myosin ATPase	L
	Sarcoplasmic reticulum Ca^{2+} -ATPase	H

[59%]

☒ D.

Condition	ATPase	Rate of ATP hydrolysis
On earth	Myosin ATPase	H
	Sarcoplasmic reticulum Ca^{2+} -ATPase	H
Spaceflight	Myosin ATPase	L
	Sarcoplasmic reticulum Ca^{2+} -ATPase	L

[25%]

Correct

25% answered correctly

Explanation:

Spaceflight reduces ATP hydrolysis in postural muscles during non-exercise activity

Condition	ATPase	Rate of ATP hydrolysis
On earth	Myosin ATPase	Higher
	Sarcoplasmic reticulum Ca^{2+} -ATPase	Higher
Spaceflight	Myosin ATPase	Lower
	Sarcoplasmic reticulum Ca^{2+} -ATPase	Lower

Because of the reduced gravity in space, less postural muscle tension development is required, thereby reducing the need for ATP hydrolysis relative to normal conditions on Earth.

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The relative **positioning** of the bones (eg, vertebrae in the spine, long bones in the limbs) in the body's various regions largely determines posture. Contraction or relaxation of postural muscles (ie, muscles controlling posture) allows posture to be maintained or adjusted, and on Earth one of the main functions of postural muscles is to oppose the pull of gravity.

There is an energetic cost associated with muscle contraction. **Hydrolysis of ATP** by myosin ATPase alters the position of the myosin head, correctly positioning it for the force-generating interaction between actin and myosin during the power stroke of the crossbridge cycle. In addition, movement of ions across cellular membranes is necessary to allow muscle activation, force development, and relaxation to occur. ATP-requiring pumps such as the plasma membrane **Na^+/K^+ -ATPase** and SERCA use the energy released during ATP hydrolysis to pump ions involved in muscle

contraction against their concentration gradients.

The question asks for the likely effect of spaceflight on postural muscle myosin

ATPase and SERCA activities during nonexercise spaceflight activities. Contractile activity requires ATP hydrolysis by both the myosin ATPase (to move the thick and thin filaments relative to one another) and SERCA (to transport Ca^{2+} across the sarcoplasmic reticulum membrane). Therefore, in the microgravity environment of space, **both myosin ATPase and SERCA activities** would **likely be lower during nonexercise spaceflight** activities than on Earth.

(Choices A, B, and C) The amount of force necessary to maintain posture is reduced in microgravity environments. As a result, less Ca^{2+} will be pumped by SERCA and less myosin ATPase activity will be necessary during the crossbridge cycle in microgravity than in Earth conditions.

Educational objective:

Muscle contractions increase hydrolysis of ATP by myosin ATPase, the sarcoplasmic reticulum Ca^{2+} -ATPase, and the $\text{Na}^{+}/\text{K}^{+}$ -ATPase.

Time spent:0 sec

Subject:

Biology

QID:404065

System:

Musculoskeletal System

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The close physical interactions involved in the initiation of skeletal muscle force production occur in what sequence?

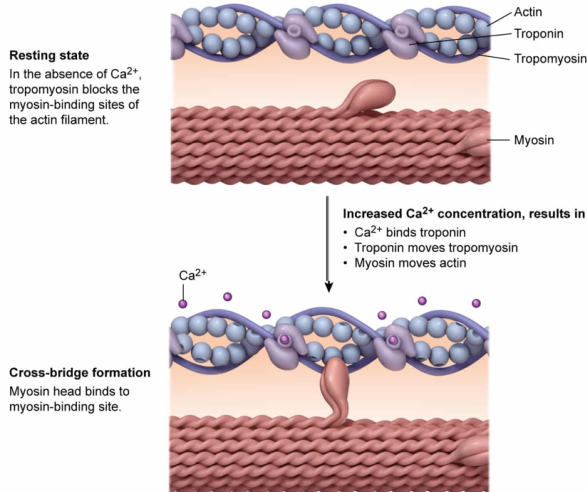
- ✔ ☒ A. Ca^{2+} binds troponin → troponin moves tropomyosin → myosin moves actin [86%]
- ☐ B. Na^+ binds tropomyosin → tropomyosin moves troponin → myosin moves actin
- ☐ C. Na^+ binds troponin → troponin moves tropomyosin → actin moves myosin [1%]
- ☐ D. Ca^{2+} binds tropomyosin → tropomyosin moves troponin → myosin moves actin [11%]

Correct

87% answered correctly

Explanation:

Increased Ca^{2+} causes troponin to shift tropomyosin's position on actin, thereby allowing strong binding with myosin



Several different proteins contribute to the initiation of **skeletal muscle** force production, including the following:

- A complex of **troponin** molecules, one of which binds Ca^{2+} , that interact with tropomyosin
- **Tropomyosin**, which wraps around actin and covers myosin-binding sites
- **Actin**, a protein polymer associated with troponin and tropomyosin that forms the thin filament of muscle fibers and binds myosin
- **Myosin**, the thick filament of muscle fibers that binds to and moves along the actin of the thin filament

The question asks for the correct sequence of physical interactions occurring during the initiation of force production in skeletal muscle. **Binding of Ca^{2+} to troponin causes troponin to shift** the position of **tropomyosin** on the actin filament. This shift **allows myosin to bind actin more strongly**, which is **a requisite for force production**. Therefore, the correct sequence of physical interactions initiating skeletal muscle force production is **Ca^{2+} binds troponin → troponin moves tropomyosin → myosin moves actin**.

(Choices B and D) Binding of Ca^{2+} (*not* Na^{+}) to *troponin* occurs before *troponin moves tropomyosin* on actin.

(Choice C) In addition to the fact that it is Ca^{2+} (*not* Na^{+}) that binds to troponin to start the sequence, it is myosin that moves actin (*not* actin that moves myosin).

Educational objective:

Binding of Ca^{2+} to troponin causes it to move tropomyosin on the actin filament, thereby allowing the stronger binding between actin and myosin that is necessary for skeletal muscle contraction to occur.

subject.
Biology

system.
Musculoskeletal System

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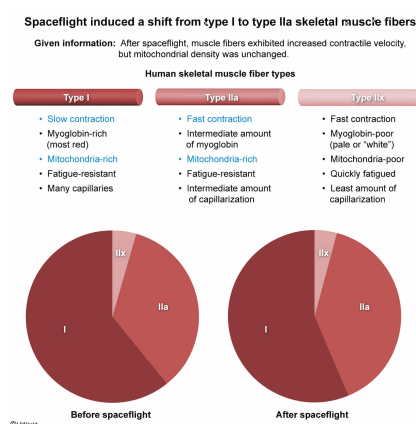
Consider an experiment in which increased contractile velocity is observed in skeletal muscle fibers from astronauts after spaceflight, with no change in mitochondrial density. These results suggest that spaceflight induces the following change of type in these muscle fibers:

- ☐ A. IIa to I [15%]
- ☐ B. IIx to I [14%]
- ✓ ☒ C. I to IIa [36%]
- ☐ D. IIa to IIx [32%]

Correct

35% answered correctly

Explanation:



Skeletal muscle fibers can be grouped into three broad categories according to their characteristics (eg, velocity of contraction, protein expression, enzyme activity) as follows:

- **Type I fibers** (slow-twitch fibers) have the **slowest** contractile velocity but are **rich in mitochondria** and capillaries. These fibers are fatigue resistant and are the most recruited fibers (in conditions on Earth). Accordingly, type I fibers are prevalent in postural muscles.
- **Type IIa** (fast-twitch, oxidative-glycolytic) fibers contract several times **faster than type I** fibers but have **similar mitochondrial density to type I** fibers. These fibers are also fatigue resistant.
- **Type IIx** (fast-twitch, glycolytic) fibers, like type IIa fibers, contract several times **faster than type I** fibers but typically have **much lower mitochondrial density** than type I or IIa fibers. Accordingly, these type IIx fibers are more reliant upon anaerobic metabolism (ie, [glycolysis](#), [fermentation](#)) than are type I and IIa fibers, which are more reliant on [aerobic respiration](#).

The question describes an experiment in which spaceflight increased the contractile velocity of astronaut muscle fiber samples but did not change the fibers' mitochondrial density. The change in contractile velocity corresponds to a shift from type I to type II fibers, and the number of mitochondria in type IIa fibers is similar to that in type I fibers. Therefore, a **type I → IIa shift** is **consistent with the response to spaceflight described** in the question.

(Choices A and B) Type I fibers are the slowest-contracting skeletal muscle fibers. In addition, type IIx fibers typically have the lowest mitochondrial density of the fiber types listed in the answer choices. Therefore, the fiber-type transitions in these choices are inconsistent with a transition that increases contractile velocity without a change in mitochondrial density.

(Choice D) Type IIa skeletal muscle fibers generally have greater mitochondrial density than type IIx fibers, not similar mitochondrial density.

Educational objective:

Skeletal muscle fibers can be grouped into broad categories according to their characteristics. Slow (type I) fibers are rich in mitochondria, fatigue-resistant, and common in postural muscles. Fast (type II) fibers contract much faster than type I fibers and have either high mitochondrial density and fatigue resistance (type IIa fibers) or relatively low mitochondrial density and fatigue resistance (type IIx).

Time spent:0 sec
Subject:
Biology

QID:404067
System:
Musculoskeletal System

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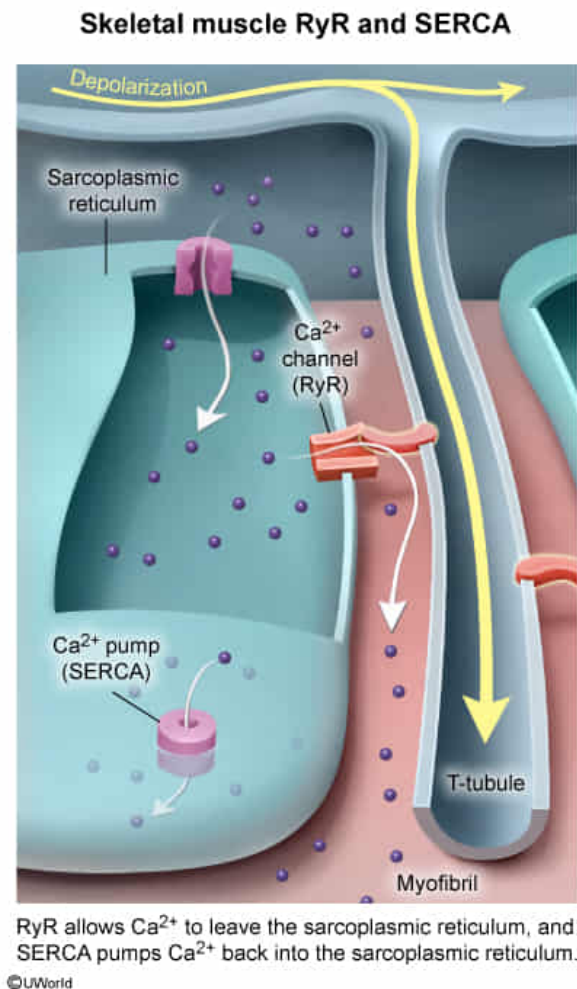
The ryanodine receptor (RyR) is a Ca^{2+} channel, and sarcoplasmic reticulum Ca^{2+} -ATPase (SERCA) is an ATPase that transports Ca^{2+} . In which direction do RyR and SERCA transport Ca^{2+} in response to depolarization of a muscle fiber?

- ✔ ☒ A. RyR transports Ca^{2+} out of the sarcoplasmic reticulum; SERCA transports Ca^{2+} into the sarcoplasmic reticulum. [42%]
- ☐ B. RyR transports Ca^{2+} into the sarcoplasmic reticulum; SERCA transports Ca^{2+} out of the sarcoplasmic reticulum. [35%]
- ☐ C. Both RyR and SERCA transport Ca^{2+} out of the sarcoplasmic reticulum. [11%]
- ☐ D. Both RyR and SERCA transport Ca^{2+} into the sarcoplasmic reticulum. [10%]

Correct

42% answered correctly

Explanation:



The **sarcoplasmic reticulum** (SR) is a specialized **endoplasmic reticulum** found in **skeletal muscle fibers**. The main role of the SR is to store calcium ions (Ca^{2+}) that, if constantly present in the cytosol, would lead to inappropriate activation of numerous processes (eg, proteolysis by Ca^{2+} -activated **proteases**). In striated muscle fibers (ie, those of cardiac and skeletal muscles), the SR is the source of most Ca^{2+} released into the cytosol in response to stimuli (eg, depolarization).

The storage of Ca^{2+} within the SR generates a large Ca^{2+} concentration gradient across the wall of the SR. Opening of the ryanodine receptor (RyR), a Ca^{2+} channel located in the SR membrane, allows a large amount of Ca^{2+} to rapidly diffuse down its concentration gradient into the cytosol. SERCA pumps the released Ca^{2+} back into the SR, restoring the initial gradient and allowing for efficient recycling of intracellular Ca^{2+} stores.

The question asks in which direction RyR and SERCA transport Ca^{2+} in response to depolarization of a muscle fiber. In **response to depolarization**, **Ca^{2+} leaves the SR via RyR** and, shortly thereafter, is **transported back into the SR by SERCA**.

(Choice B) Ca^{2+} diffuses *out* of the SR via RyR and is transported *into* the SR via SERCA.

(Choices C and D) Only RyR transports Ca^{2+} out of the SR, and only SERCA

transports Ca^{2+} back into the SR.

Educational objective:

The sarcoplasmic reticulum (SR) stores the Ca^{2+} necessary for skeletal muscle contraction to occur. A specific Ca^{2+} channel permits diffusion of Ca^{2+} out of the SR, and a specific ATPase pumps Ca^{2+} from the cytosol back into the SR.

Time spent:0 sec

Subject:

Biology

QID:404068

System:

Musculoskeletal System

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The purpose of the bed rest study described in the passage was:

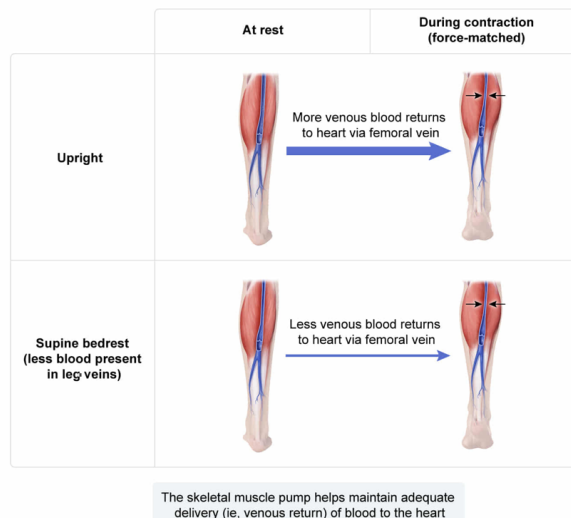
- ☐ A. to determine whether muscle contraction can modulate femoral artery blood flow. [3%]
- ☐ B. to determine whether muscle contraction can modulate femoral vein blood flow. [53%]
- ☒ C. to determine whether venous blood volume can modulate the efficiency by which blood is pumped to the heart by skeletal muscle contraction. [24%]
- ☐ D. to determine whether the efficiency of blood pumping to the heart by skeletal muscle contraction modulates skeletal muscle force production. [18%]

Correct

25% answered correctly

Explanation:

Bedrest reduces the efficiency of blood delivery to the heart due to skeletal muscle contraction (ie, the “skeletal muscle pump”)



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In a steady state (eg, at rest), the volumes of blood **entering and leaving** the heart are equal, as are the volumes of blood entering and leaving each chamber. The distribution of the blood pumped into the **systemic circulation**, however, can vary considerably according to the influence of several factors. For example, in response to changing environmental temperatures, blood is **redistributed** to maintain core body **temperature homeostasis**.

The passage describes an experiment in which blood flow in the **femoral vein** is measured during upright posture as well as during bed rest, and the question asks for the purpose of the study. According to the passage, the **researchers controlled** the amount of **force produced by the muscle contraction**, and the **intervention** would be expected to **alter** the amount of **blood distributed to the leg veins**. Therefore, by measuring femoral vein blood flow in response to controlled contractions, the researchers could **determine** whether the volume of blood in the veins modulates the **efficiency** of blood pumped to the heart by skeletal muscle contraction.

(Choice A) Although femoral artery and femoral vein blood flow will be similar under most circumstances, femoral *artery* blood flow was not measured in the research study described in the passage. Therefore, the purpose of the study was *not* to determine the effect of muscle contraction on femoral artery blood flow.

(Choice B) Had the researchers been solely interested in the effect of contraction on femoral venous blood flow, it would have been unnecessary to include testing in two body positions (ie, upright, bed rest). This makes it unlikely that the purpose of the study was simply to determine the effect of muscle contraction on femoral vein blood flow.

(Choice D) Skeletal muscle force production was included as a **control variable** in the design of the study. Because the researchers set the force produced by the calf muscles at a predetermined level, the purpose of the study was not to determine

whether muscle pump efficiency modulates skeletal muscle force production.

Educational objective:

Skeletal muscle contraction can compress veins and thereby increase the delivery of venous blood to the heart.

Time spent:0 sec

Subject:
Biology

QID:404069

System:
Musculoskeletal System

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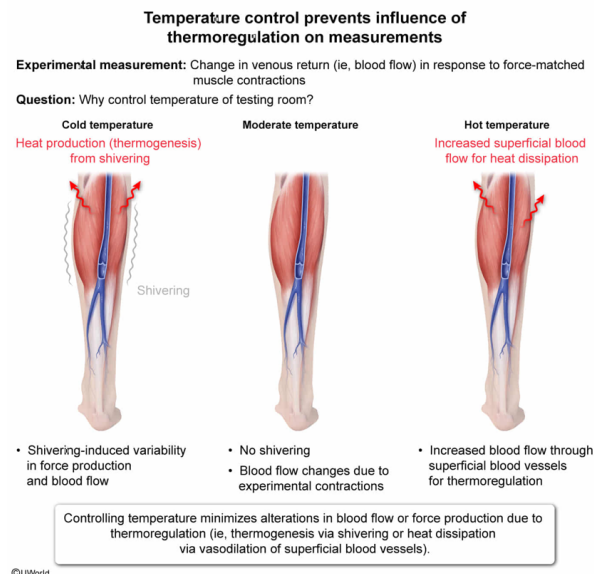
What is a likely reason the researchers controlled the environmental temperature during the study described in the passage?

- ☐ A. To determine whether temperature affects the return of venous blood from the skin to the heart
- ☐ B. To determine the effect of temperature on the return of venous blood from the lower leg muscles to the heart [1%]
- ☐ C. To reduce variability in blood flow measurements by ensuring that baseline constriction of leg veins was maximal [28%]
- ☒ D. To minimize the likelihood that variable levels of skeletal muscle thermogenesis affected the study results [69%]

Correct

69% answered correctly

Explanation:



In addition to its ability to markedly increase its tension, **skeletal muscle** also possesses the ability to greatly increase its rate of energy expenditure. During vigorous exercise, increased energy expenditure by skeletal muscle is the primary contributor to the increased rate of whole-body energy expenditure, which can greatly increase relative to baseline rates in some individuals. In addition, cold exposure can induce rapid, involuntary muscle contractions (ie, shivering) that can also markedly increase whole-body energy expenditure.

The mobilization of the energy required for muscle contraction is **not perfectly efficient**; more energy is converted to heat than is used to perform the contraction. As a result, increasing muscle energy expenditure via exercise or shivering markedly increases the generation of heat, which can help maintain **body temperature homeostasis** during cold exposure.

The passage describes an experiment in which blood flow responses to calf muscle contraction were measured, and the question asks why the researchers controlled environmental temperature during the experiment. In addition to influencing the relative distribution of blood between the body's core and surface, the environmental temperature can, when low enough, also stimulate thermogenesis (ie, heat generation) by shivering.

Thermogenesis from shivering would likely add baseline variability to the blood flow measurements due to involuntary contractions that compress leg veins and increase demand for blood flow to the shivering muscles. Therefore, by controlling the environmental temperature the researchers could prevent thermogenesis from influencing the results.

(Choices A and B) The passage describes the environmental temperature as controlled and not experimentally *varied*. Because the environmental temperature was experimentally designed *not* to vary, the purpose of the experimental control of temperature was *not* to compare blood flow responses at different temperatures.

(Choice C) If temperature were controlled in a way that maximized leg vein constriction, the temperature would be cold. A low temperature would increase the likelihood of shivering, which would likely *add* baseline variability to the blood flow measurements due to involuntary contractions that compress leg veins.

Educational objective:

One function of muscle is to increase heat production (thermogenesis) by shivering to maintain or increase body temperature.

Time spent:0 sec

Subject:
Biology

QID:404070

System:
Musculoskeletal System